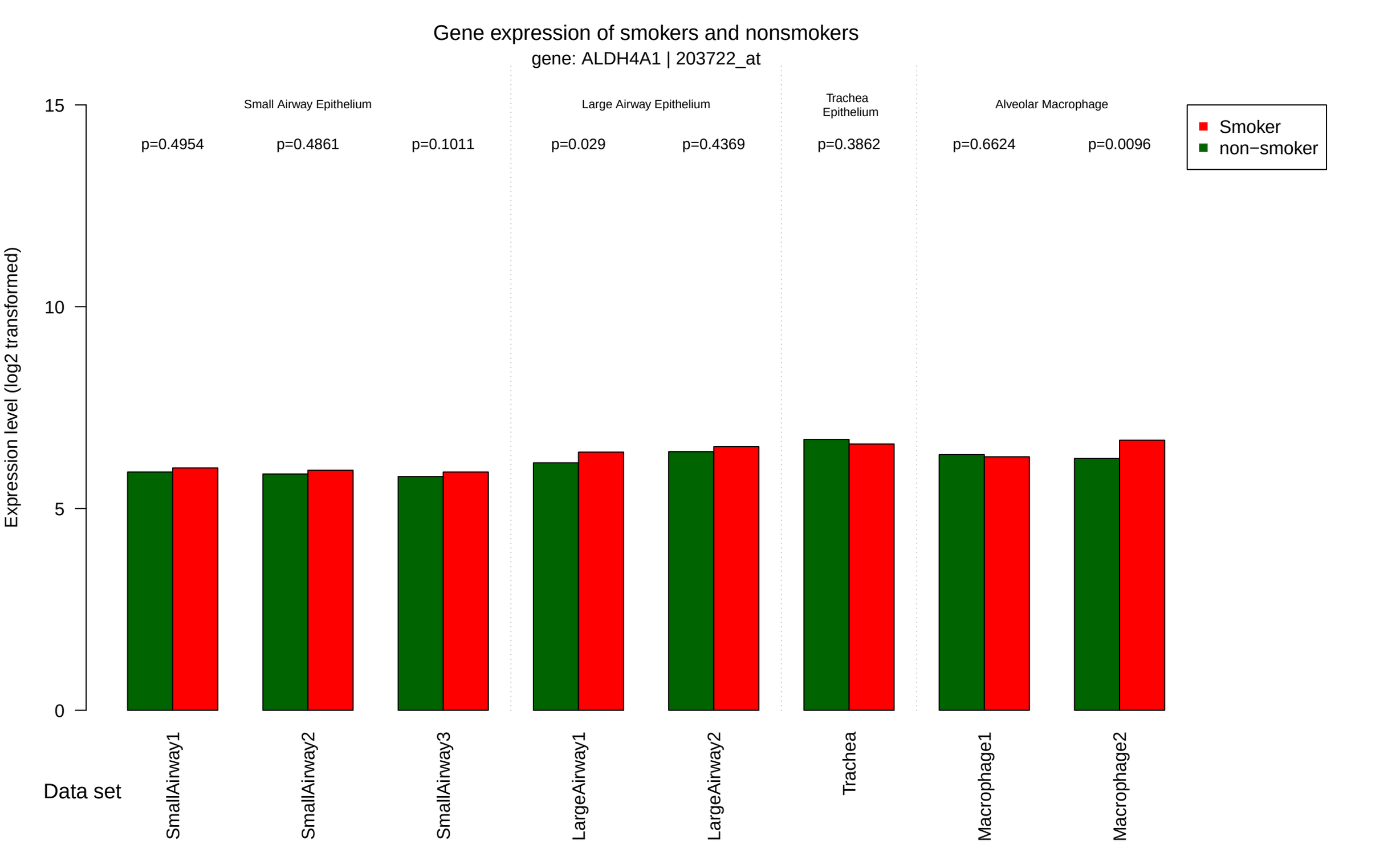
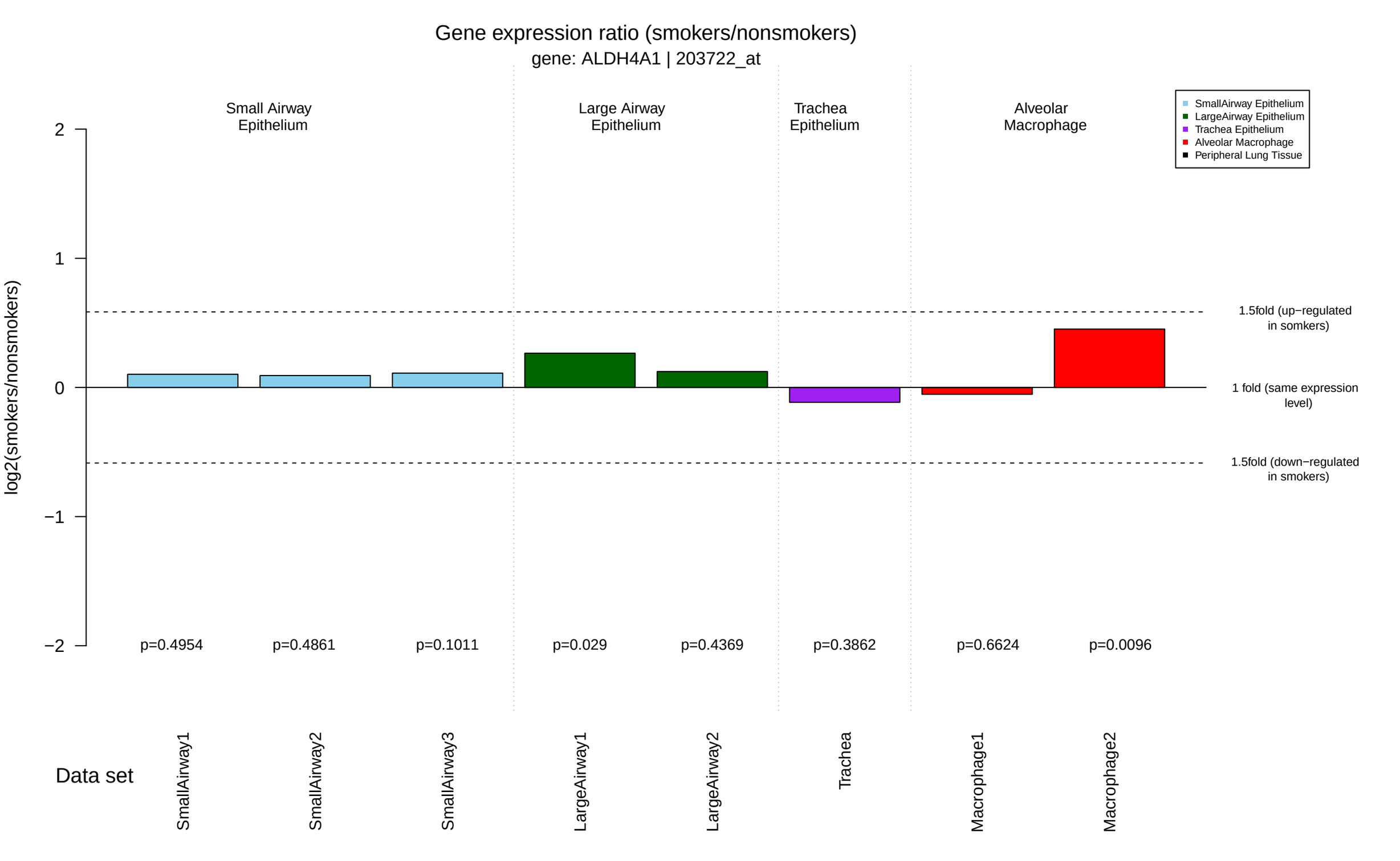
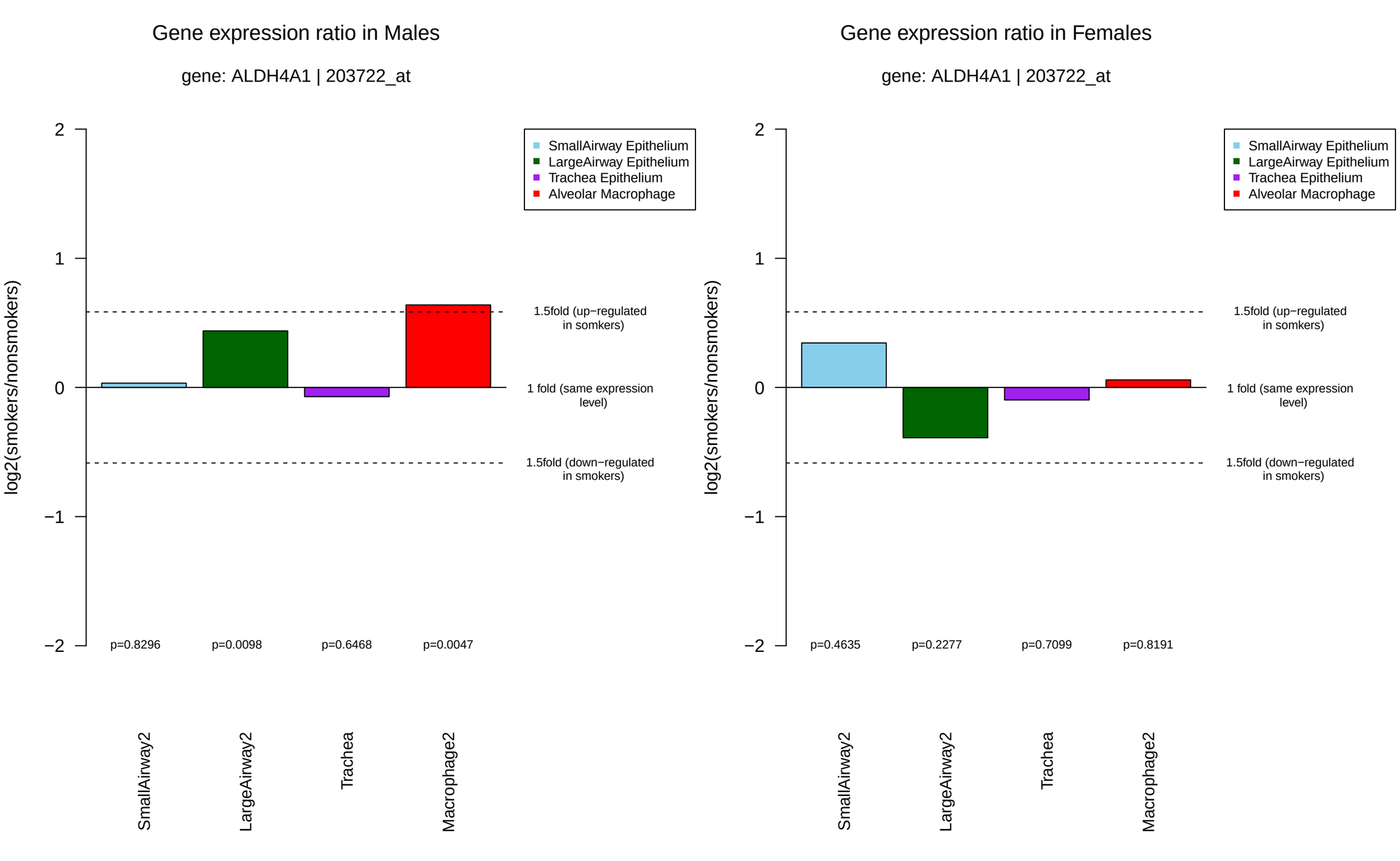
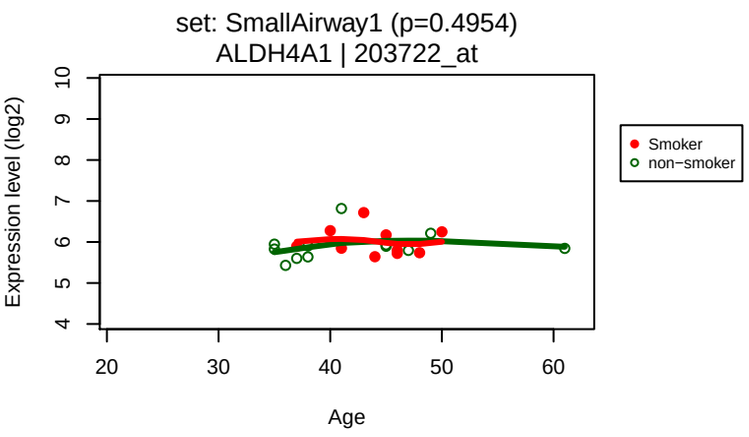
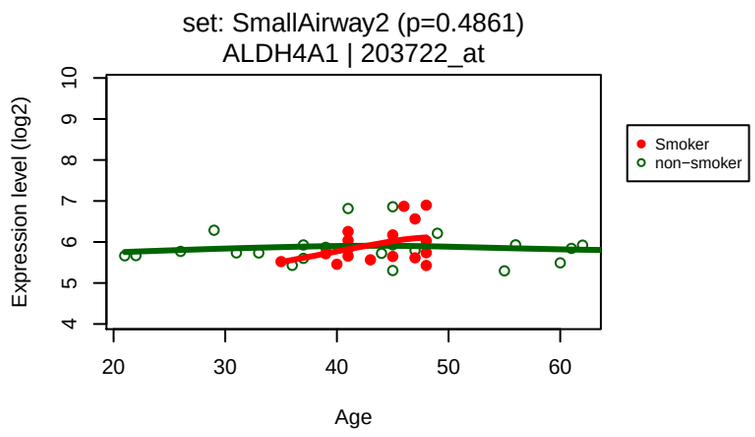
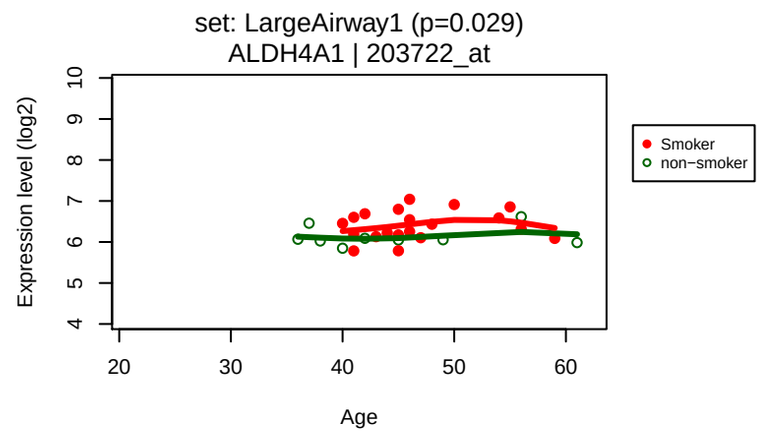

Gene expression of smokers and nonsmokers
gene: ALDH4A1 | 203722_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4954
p=0.4861
p=0.1011
p=0.029
p=0.4369
p=0.3862
p=0.6624
p=0.0096
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

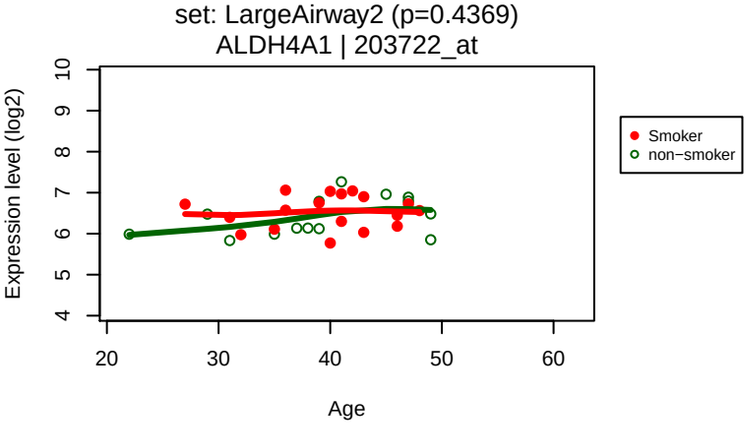

Gene expression ratio (smokers/nonsmokers)
gene: ALDH4A1 | 203722_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4954
p=0.4861
p=0.1011
p=0.029
p=0.4369
p=0.3862
p=0.6624
p=0.0096
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

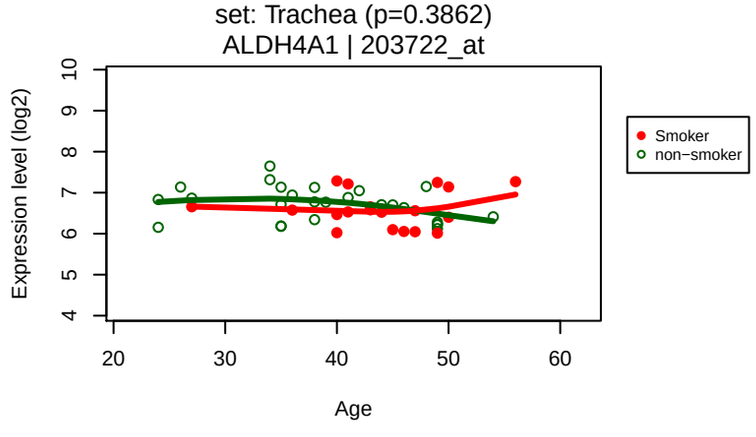

Gene expression ratio in Males
gene: ALDH4A1 | 203722_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8296
p=0.0098
p=0.6468
p=0.0047
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH4A1 | 203722_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4635
p=0.2277
p=0.7099
p=0.8191
SmallAirway2
LargeAirway2
Trachea
Macrophage2

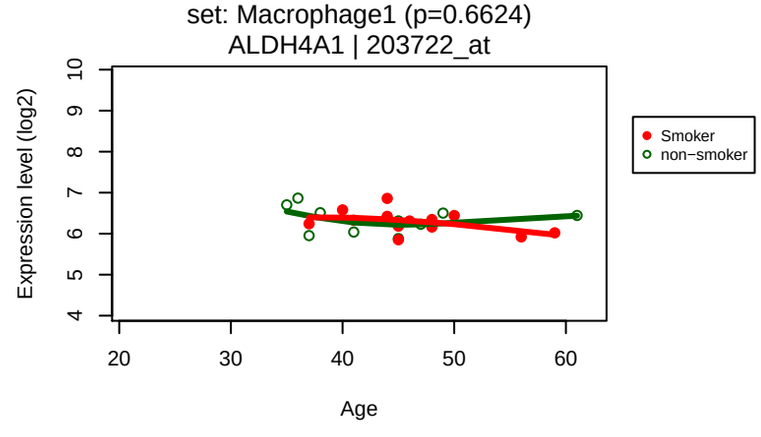

set: SmallAirway1 (p=0.4954)
ALDH4A1 | 203722_at
Expression level (log2)
Age
Smoker
non−smoker

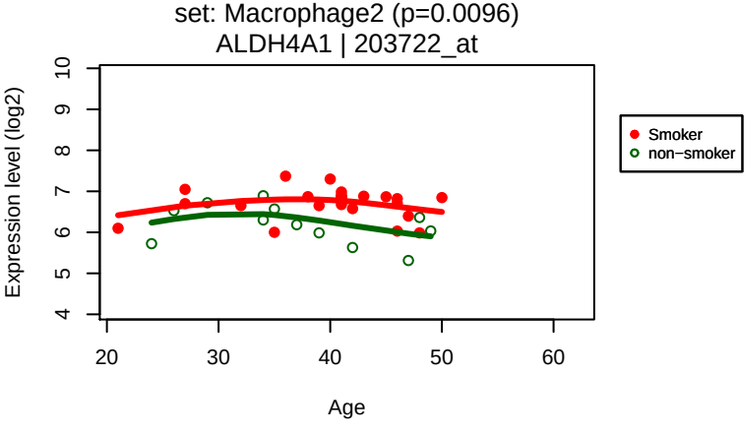

set: SmallAirway2 (p=0.4861)
ALDH4A1 | 203722_at
Expression level (log2)
Age
Smoker
non−smoker

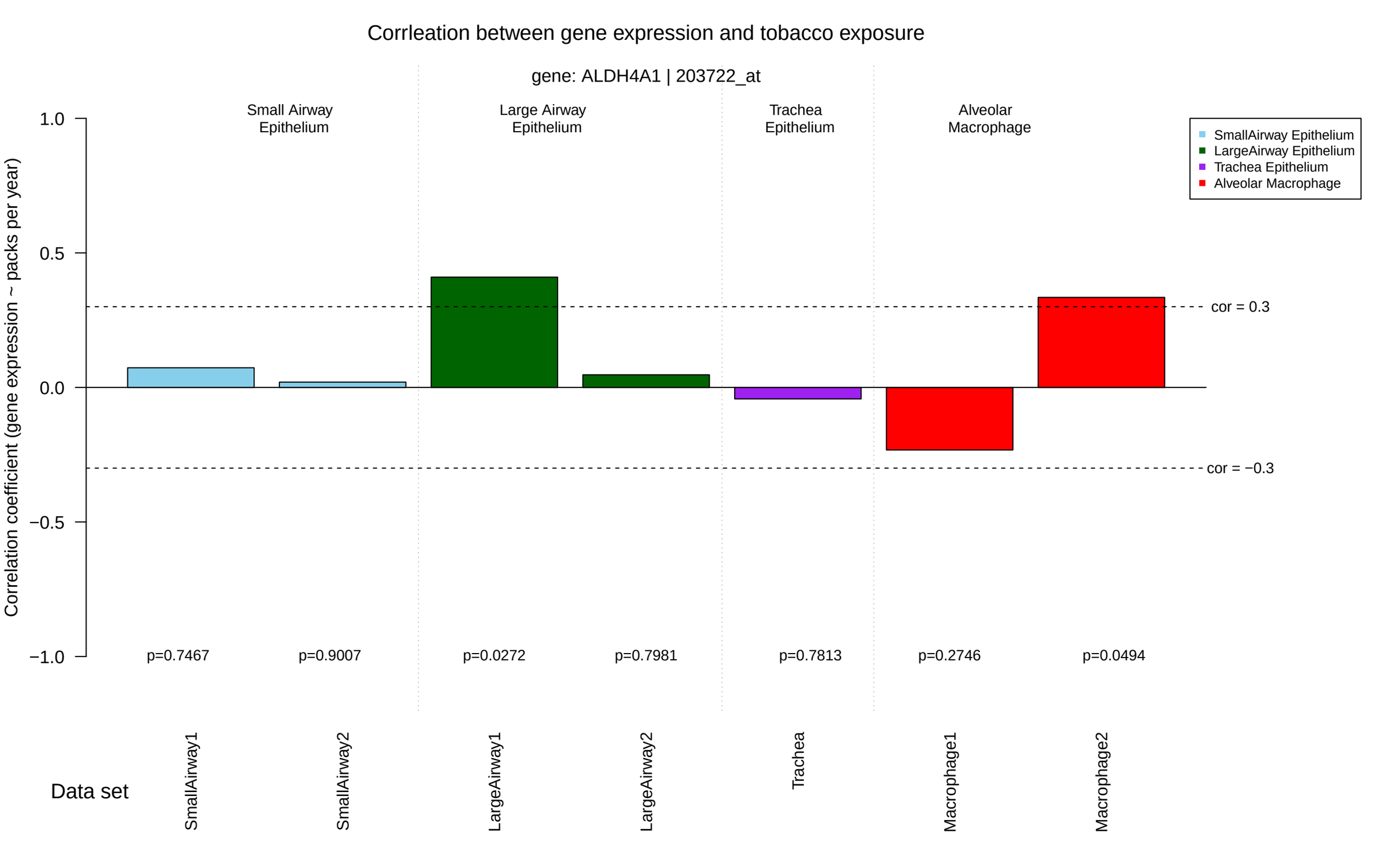

set: LargeAirway1 (p=0.029)
ALDH4A1 | 203722_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4369)
ALDH4A1 | 203722_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3862)
ALDH4A1 | 203722_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6624)
ALDH4A1 | 203722_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0096)
ALDH4A1 | 203722_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH4A1 | 203722_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7467
p=0.9007
p=0.0272
p=0.7981
p=0.7813
p=0.2746
p=0.0494
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2